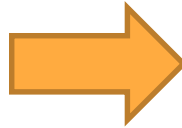

Let's Practice

**IF
Statement**

Example 1

```
always @ (*)
begin
  if( done_flag == 2'b00)
    begin
      A = 1'b1 ;
      B = 1'b0 ;
    end
  else if ( done_flag == 2'b01)
    begin
      A = 1'b0 ;
      B = 1'b1 ;
    end
  else
    begin
      A = 1'b1 ;
      B = 1'b0 ;
    end
  end
end
```

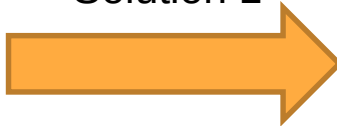


No Unintentional Latch will be inferred

Example 2

```
always @ (*)
begin
  if(E)
    begin
      B = 1'b0 ;
      C = 1'b0 ;
    end
  else if (F)
    begin
      A = 1'b0 ;
      B = 1'b1 ;
      C = 1'b1 ;
    end
  else if (G)
    begin
      A = 1'b0 ;
    end
  else
    begin
      A = 1'b1 ;
      B = 1'b0 ;
    end
end
```

Solution 1



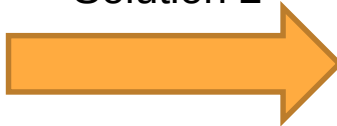
```
always @ (*)
begin
  if(E)
    begin
      A = 1'b0 ;
      B = 1'b0 ;
      C = 1'b0 ;
    end
  else if (F)
    begin
      A = 1'b0 ;
      B = 1'b1 ;
      C = 1'b1 ;
    end
end
```

```
else if (G)
  begin
    A = 1'b0 ;
    B = 1'b0 ;
    C = 1'b0 ;
  end
else
  begin
    A = 1'b1 ;
    B = 1'b0 ;
    C = 1'b0 ;
  end
end
```

Example 2

```
always @ (*)
begin
  if(E)
    begin
      B = 1'b0 ;
      C = 1'b0 ;
    end
  else if (F)
    begin
      A = 1'b0 ;
      B = 1'b1 ;
      C = 1'b1 ;
    end
  else if (G)
    begin
      A = 1'b0 ;
    end
  else
    begin
      A = 1'b1 ;
      B = 1'b0 ;
    end
end
```

Solution 2



```
always @ (*)
begin
  A = 1'b0 ;
  B = 1'b0 ;
  C = 1'b0 ;
  if(E)
    begin
      B = 1'b0 ;
      C = 1'b0 ;
    end
  else if (F)
    begin
      A = 1'b0 ;
      B = 1'b1 ;
      C = 1'b1 ;
    end
end
```

```
else if (G)
  begin
    A = 1'b0 ;
  end
else
  begin
    A = 1'b1 ;
    B = 1'b0 ;
  end
end
```

Example 3

```
always @ (*)
begin
  if( done_flag == 2'b00)
    begin
      A = 1'b1 ;
      B = 1'b0 ;
    end
  else if ( done_flag == 2'b01)
    begin
      A = 1'b0 ;
      B = 1'b1 ;
    end
  else if ( done_flag == 2'b11)
    begin
      A = 1'b1 ;
      B = 1'b0 ;
    end
end
```

Solution



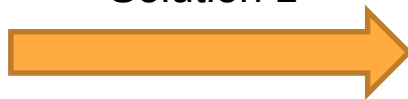
```
always @ (*)
begin
  if( done_flag == 2'b00)
    begin
      A = 1'b1 ;
      B = 1'b0 ;
    end
  else if ( done_flag == 2'b01)
    begin
      A = 1'b0 ;
      B = 1'b1 ;
    end
  else if ( done_flag == 2'b11)
    begin
      A = 1'b1 ;
      B = 1'b0 ;
    end
end
```

```
else
  begin
    A = 1'b0 ;
    B = 1'b0 ;
  end
end
```

Example 4

```
always @ (*)
begin
  if( done_flag == 2'b00)
    begin
      A = 1'b1 ;
      B = 1'b0 ;
      C = 1'b0 ;
    end
  else if ( done_flag == 2'b01)
    begin
      A = 1'b0 ;
      B = 1'b1 ;
      C = 1'b0 ;
    end
  else
    begin
      A = 1'b1 ;
      B = 1'b0 ;
    end
end
```

Solution 1

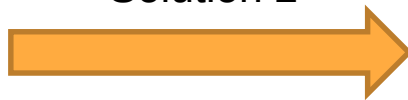


```
always @ (*)
begin
  if( done_flag == 2'b00)
    begin
      A = 1'b1 ;
      B = 1'b0 ;
      C = 1'b0 ;
    end
  else if ( done_flag == 2'b01)
    begin
      A = 1'b0 ;
      B = 1'b1 ;
      C = 1'b0 ;
    end
  else
    begin
      A = 1'b1 ;
      B = 1'b0 ;
      C = 1'b0 ;
    end
end
```

Example 4

```
always @ (*)
begin
  if( done_flag == 2'b00)
    begin
      A = 1'b1 ;
      B = 1'b0 ;
      C = 1'b0 ;
    end
  else if ( done_flag == 2'b01)
    begin
      A = 1'b0 ;
      B = 1'b1 ;
      C = 1'b0 ;
    end
  else
    begin
      A = 1'b1 ;
      B = 1'b0 ;
    end
end
```

Solution 2



```
always @ (*)
begin
  C = 1'b0 ;
  if( done_flag == 2'b00)
    begin
      A = 1'b1 ;
      B = 1'b0 ;
      C = 1'b0 ;
    end
  else if ( done_flag == 2'b01)
    begin
      A = 1'b0 ;
      B = 1'b1 ;
      C = 1'b0 ;
    end
  else
    begin
      A = 1'b1 ;
      B = 1'b0 ;
    end
end
```

Let's Practice

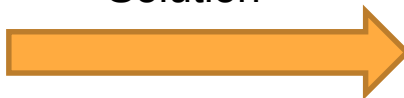
Case Statement



Example 1

```
always @ (*)
case (sel)
  2'b00: begin
    A = 1'b1 ;
    B = 1'b0 ;
    C = 1'b1 ;
  end
  2'b01: begin
    A = 1'b1 ;
    B = 1'b0 ;
    C = 1'b1 ;
  end
  2'b10: begin
    A = 1'b1 ;
    B = 1'b0 ;
    C = 1'b1 ;
  end
endcase
```

Solution

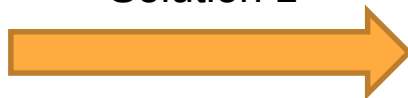


```
always @ (*)
case (sel)
  2'b00: begin
    A = 1'b1 ;
    B = 1'b0 ;
    C = 1'b1 ;
  end
  2'b01: begin
    A = 1'b1 ;
    B = 1'b0 ;
    C = 1'b1 ;
  end
  2'b10: begin
    A = 1'b1 ;
    B = 1'b0 ;
    C = 1'b1 ;
  end
  default: begin
    A = 1'b1 ;
    B = 1'b0 ;
    C = 1'b1 ;
  end
endcase
```

Example 2

```
always @ (*)
case (sel)
  2'b00: begin
    A = 1'b1 ;
    B = 1'b0 ;
    C = 1'b1 ;
  end
  2'b01: begin
    B = 1'b1 ;
  end
  2'b10: begin
    A = 1'b1 ;
  end
  2'b11: begin
    A = 1'b1 ;
    B = 1'b0 ;
  end
endcase
```

Solution 1



```
always @ (*)
case (sel)
  2'b00: begin
    A = 1'b1 ;
    B = 1'b0 ;
    C = 1'b1 ;
  end
  2'b01: begin
    A = 1'b1 ;
    C = 1'b0 ;
    B = 1'b1 ;
  end
  2'b10: begin
    A = 1'b1 ;
    B = 1'b0 ;
    C = 1'b1 ;
  end
  2'b11: begin
    A = 1'b1 ;
    B = 1'b0 ;
    C = 1'b1 ;
  end
endcase
```

Example 2

```
always @ (*)
case (sel)
  2'b00: begin
    A = 1'b1 ;
    B = 1'b0 ;
    C = 1'b1 ;
  end
  2'b01: begin
    B = 1'b1 ;
  end
  2'b10: begin
    A = 1'b1 ;
  end
  2'b11: begin
    A = 1'b1 ;
    B = 1'b0 ;
  end
endcase
```

Solution 2



```
always @ (*)
begin
  A = 1'b1 ;
  B = 1'b0 ;
  C = 1'b1 ;
  case (sel)
    2'b00: begin
      A = 1'b1 ;
      B = 1'b0 ;
      C = 1'b1 ;
    end
    2'b01: begin
      B = 1'b1 ;
    end
    2'b10 : begin
      A = 1'b1 ;
    end
    2'b11 : begin
      A = 1'b1 ;
      B = 1'b0 ;
    end
  endcase
end
```

Example 3

```
always @ (*)
begin
    A = 1'b1 ;
    B = 1'b0 ;
    case (sel)
        2'b00: begin
            A = 1'b1 ;
            B = 1'b0 ;
            C = 1'b1 ;
        end
        2'b01: begin
            B = 1'b1 ;
        end
        2'b10: begin
            A = 1'b1 ;
        end
        default: begin
            A = 1'b1 ;
            B = 1'b0 ;
        end
    endcase
end
```

Solution 1

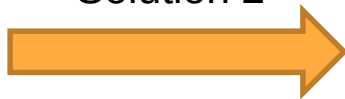


```
always @ (*)
begin
    A = 1'b1 ;
    B = 1'b0 ;
    C = 1'b1 ;
    case (sel)
        2'b00: begin
            A = 1'b1 ;
            B = 1'b0 ;
            C = 1'b1 ;
        end
        2'b01: begin
            B = 1'b1 ;
        end
        2'b10: begin
            A = 1'b1 ;
        end
        default: begin
            A = 1'b1 ;
            B = 1'b0 ;
        end
    endcase
end
```

Example 3

```
always @ (*)
begin
    A = 1'b1 ;
    B = 1'b0 ;
    case (sel)
        2'b00: begin
            A = 1'b1 ;
            B = 1'b0 ;
            C = 1'b1 ;
        end
        2'b01: begin
            B = 1'b1 ;
        end
        2'b10: begin
            A = 1'b1 ;
        end
        default: begin
            A = 1'b1 ;
            B = 1'b0 ;
        end
    endcase
end
```

Solution 2



```
always @ (*)
begin
    A = 1'b1 ;
    B = 1'b0 ;
    case (sel)
        2'b00: begin
            A = 1'b1 ;
            B = 1'b0 ;
            C = 1'b1 ;
        end
        2'b01: begin
            B = 1'b1 ;
            C = 1'b1 ;
        end
        2'b10: begin
            A = 1'b1 ;
            C = 1'b1 ;
        end
        default: begin
            A = 1'b1 ;
            C = 1'b1 ;
        end
    endcase
end
```

Example 4

```
always @ (*)
begin
    A = 1'b1 ;
    B = 1'b0 ;
    C = 1'b1 ;
case (sel)
    2'b00: begin
        A = 1'b1 ;
    end
    2'b01: begin
        B = 1'b1 ;
    end
    2'b10: begin
        C = 1'b1 ;
    end
    default: begin
        A = 1'b1 ;
        B = 1'b0 ;
    end
endcase
end
```



No Unintentional Latch will be inferred